

Technical Documentation: Interactive Product Catalog (Flutter)

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GitHub Repository: <https://github.com/SudarshanDevadiga/Dynamic-Product-Catalog>

1. Project Overview

This application serves as a practical implementation of reactive state management and modular UI design in Flutter. It features a dynamic product catalog that allows users to seamlessly browse, filter by category, and search via text input. The project emphasizes a clean separation between the data model and the user interface, utilizing `setState` for real-time visual updates and Material 3 design principles for a modern aesthetic.

2. Directory Structure & Architecture

The codebase is intentionally partitioned to separate structural layout from data definitions, ensuring maintainability:

- **lib/product.dart:** Acts as the data layer. It defines the core data structures and hosts the static mock data used to populate the application.
- **lib/main.dart:** Acts as the presentation layer. It contains the application entry point, global theme configurations, and all interactive stateful UI components.

3. Core Components & Logic

Data Layer (product.dart)

- **Product Class:** The blueprint for individual items, strongly typed with properties for `id`, `name`, `category`, `price (INR)`, `icon`, and `themeColor`.
- **sampleProducts:** A constant array acting as an in-memory database, supplying diverse items across categories like Electronics, Apparel, and Footwear to test the filtering mechanics.

```

lib > product.dart
1  import 'package:flutter/material.dart';
2
3  class Product {
4    final String id;
5    final String name;
6    final String category;
7    final double price; // in INR
8    final IconData icon;
9    final Color themeColor;
10
11   const Product({
12     required this.id,
13     required this.name,
14     required this.category,
15     required this.price,
16     required this.icon,
17     required this.themeColor,
18   });
19 }
20
21 // In-memory sample dataset
22 final List<Product> sampleProducts = [
23   const Product(
24     id: '1',
25     name: 'Wireless Headphones',
26     category: 'Electronics',
27     price: 2999.00,
28     icon: Icons.headphones_rounded,
29     themeColor: Color(0xFF6C5CE7),
30   ),
31   const Product(
32     id: '2',
33     name: 'Running Shoes',
34     category: 'Footwear',
35     price: 4499.00,
36     icon: Icons.directions_run_rounded,
37     themeColor: Color(0xFFFF7675),
38   ),
39   const Product(
40     id: '3',
41     name: 'Mechanical Keyboard',
42     category: 'Electronics',
43     price: 9999.00,
44     icon: Icons.keyboard_rounded,

```

```

45     themeColor: Color(0xFF00CEC9),
46   ),
47   const Product(
48     id: '4',
49     name: 'Cotton T-Shirt',
50     category: 'Apparel',
51     price: 1299.00,
52     icon: Icons.checkroom_rounded,
53     themeColor: Color(0xFF74B9FF),
54   ),
55   const Product(
56     id: '5',
57     name: 'Smart Watch',
58     category: 'Electronics',
59     price: 15499.00,
60     icon: Icons.watch_rounded,
61     themeColor: Color(0xFFE17055),
62   ),
63   const Product(
64     id: '6',
65     name: 'Leather Jacket',
66     category: 'Apparel',
67     price: 18999.00,
68     icon: Icons.dry_cleaning_rounded,
69     themeColor: Color(0xFF2D3436),
70   ),
71   const Product(
72     id: '7',
73     name: 'Travel Backpack',
74     category: 'Accessories',
75     price: 3299.00,
76     icon: Icons.backpack_rounded,
77     themeColor: Color(0xFF00B894),
78   ),
79   const Product(
80     id: '8',
81     name: 'Ceramic Mug',
82     category: 'Home',
83     price: 450.00,
84     icon: Icons.coffee_rounded,
85     themeColor: Color(0xFFE84393),
86   ),
87 ];

```

Presentation Layer (main.dart)

- Initialization:** The app boots via MyApp (a StatelessWidget), applying a global MaterialApp configuration with a customized scaffoldBackgroundColor and Material 3 enabled.

- **State Management:** Driven by `_ProductListingScreenState`. It maintains the source of truth for user inputs:
 - `_searchController`: Tracks text input.
 - `_selectedCategory`: Tracks the active category pill.
 - `_filteredProducts`: The derived array of products currently matching both the text and category criteria.
- **Filtering Engine:** The `_filterProducts()` method recalculates the `_filteredProducts` list by running a cross-check between the search string and the selected category, wrapping the update in `setState()` to trigger a UI rebuild.

```
lib > main.dart
1 import 'package:flutter/material.dart';
2 import 'product.dart';
3
4 void main() {
5   runApp(const MyApp());
6 }
7
8 class MyApp extends StatelessWidget {
9   const MyApp({super.key});
10
11   @override
12   Widget build(BuildContext context) {
13     return MaterialApp(
14       debugShowCheckedModeBanner: false,
15       title: 'Modern Store',
16       theme: ThemeData(
17         useMaterial3: true,
18         scaffoldBackgroundColor: const Color(0xFFB8F9FA),
19         fontFamily: 'Roboto',
20       ),
21       home: const ProductListingScreen(),
22     );
23   }
24 }
25
26 class ProductListingScreen extends StatefulWidget {
27   const ProductListingScreen({super.key});
28
29   @override
30   State<ProductListingScreen> createState() => _ProductListingScreenState();
31 }
32
33 class _ProductListingScreenState extends State<ProductListingScreen> {
34   final TextEditingController _searchController = TextEditingController();
35   List<Product> _filteredProducts = [];
36   String _selectedCategory = 'All';
37
38   late final List<String> _categories = [
39     'All',
40     ...{for (var p in sampleProducts) p.category}
41   ];
42
43   @override
44   void initState() {
45     super.initState();
46     _filteredProducts = List.from(sampleProducts);
47   }
48
49   @override
50   void dispose() {
```

```

51     _searchController.dispose();
52     super.dispose();
53 }
54
55 void _filterProducts() {
56   final query = _searchController.text.toLowerCase();
57   setState(() {
58     _filteredProducts = sampleProducts.where((product) {
59       final matchesSearch = product.name.toLowerCase().contains(query);
60       final matchesCategory =
61         _selectedCategory == 'All' || product.category == _selectedCategory;
62
63       return matchesSearch && matchesCategory;
64     }).toList();
65   });
66 }
67
68 @override
69 Widget build(BuildContext context) {
70   return Scaffold(
71     body: SafeArea(
72       child: Column(
73         crossAxisAlignment: CrossAxisAlignment.start,
74         children: [
75           // Custom Header
76           const Padding(
77             padding: EdgeInsets.fromLTRB(20, 20, 20, 10),
78             child: Column(
79               crossAxisAlignment: CrossAxisAlignment.start,
80               children: [
81                 Text(
82                   'Explore Products',
83                   style: TextStyle(
84                     fontSize: 28,
85                     fontWeight: FontWeight.bold,
86                     color: Color(0xFF2D3436),
87                     letterSpacing: -0.5,
88                   ),
89                 ),
90                 SizedBox(height: 4),
91                 Text(
92                   'Find what you need today',
93                   style: TextStyle(
94                     fontSize: 14,
95                     color: Colors.grey,
96                   ),
97                 ),
98               ],
99             ),
100           ),

```

```

101
102     // Styled Search Input
103     Padding(
104       padding: const EdgeInsets.symmetric(horizontal: 20, vertical: 8),
105       child: Container(
106         decoration: BoxDecoration(
107           color: Colors.white,
108           borderRadius: BorderRadius.circular(16),
109           boxShadow: [
110             BoxShadow(
111               color: Colors.black.withOpacity(0.04),
112               blurRadius: 12,
113               offset: const Offset(0, 4),
114             ),
115           ],
116         ),
117         child: TextField(
118           controller: _searchController,
119           onChanged: (_) => _filterProducts(),
120           decoration: InputDecoration(
121             hintText: 'Search electronics, apparel...',
122             hintStyle:
123               TextStyle(color: Colors.grey.shade400, fontSize: 15),
124             prefixIcon:
125               const Icon(Icons.search, color: Color(0xFF6C5CE7)),
126             suffixIcon: _searchController.text.isNotEmpty
127               ? IconButton(
128                 icon: const Icon(Icons.cancel, color: Colors.grey),
129                 onPressed: () {
130                   _searchController.clear();
131                   _filterProducts();
132                 },
133               )
134               : null,
135             border: InputBorder.none,
136             contentPadding: const EdgeInsets.symmetric(
137               horizontal: 16, vertical: 14),
138           ),
139         ),
140       ),
141     ),
142
143     const SizedBox(height: 12),
144
145     // Animated Horizontal Category Chips
146     SizedBox(
147       height: 38,
148       child: ListView.builder(
149         scrollDirection: Axis.horizontal,
150         padding: const EdgeInsets.symmetric(horizontal: 16),

```

```

151     itemCount: _categories.length,
152     itemBuilder: (context, index) {
153       final category = _categories[index];
154       final isSelected = category == _selectedCategory;
155
156       return GestureDetector(
157         onTap: () {
158           setState(() {
159             _selectedCategory = category;
160           });
161           _filterProducts();
162         },
163         child: AnimatedContainer(
164           duration: const Duration(milliseconds: 200),
165           margin: const EdgeInsets.only(right: 8),
166           padding: const EdgeInsets.symmetric(horizontal: 18),
167           alignment: Alignment.center,
168           decoration: BoxDecoration(
169             color: isSelected
170               ? const Color(0xFF6CCE7)
171               : Colors.white,
172             borderRadius: BorderRadius.circular(20),
173             border: Border.all(
174               color: isSelected
175                 ? const Color(0xFF6CCE7)
176                 : Colors.grey.shade300,
177             ),
178             boxShadow: isSelected
179               ? [
180                 BoxShadow(
181                   color: const Color(0xFF6CCE7)
182                     .withOpacity(0.3),
183                   blurRadius: 8,
184                   offset: const Offset(0, 3),
185                 ),
186               ]
187               : null,
188           ),
189           child: Text(
190             category,
191             style: TextStyle(
192               color: isSelected
193                 ? Colors.white
194                 : const Color(0xFF2D3436),
195               fontWeight:
196                 isSelected ? FontWeight.bold : FontWeight.w500,
197               fontSize: 13,
198             ),
199           ),
200         ),

```

```

201       ),
202     ),
203   ),
204 ),
205
206 const SizedBox(height: 16),
207
208 // Product List
209 Expanded(
210   child: _filteredProducts.isEmpty
211     ? _buildEmptyState()
212     : ListView.builder(
213       physics: const BouncingScrollPhysics(),
214       padding: const EdgeInsets.symmetric(horizontal: 20),
215       itemCount: _filteredProducts.length,
216       itemBuilder: (context, index) {
217         final product = _filteredProducts[index];
218         return _buildProductCard(product);
219       },
220     ),
221 ),
222 ),
223 ),
224 );
225
226 // Modern Card Component with ð Symbol
227
228 Widget _buildProductCard(Product product) {
229   return Container(
230     margin: const EdgeInsets.only(bottom: 14),
231     decoration: BoxDecoration(
232       color: Colors.white,
233       borderRadius: BorderRadius.circular(20),
234       boxShadow: [
235         BoxShadow(
236           color: Colors.black.withOpacity(0.03),
237           blurRadius: 10,
238           offset: const Offset(0, 4),
239         ),
240       ],
241     ),
242     child: Padding(
243       padding: const EdgeInsets.all(12),
244       child: Row(
245         children: [
246           // Product Icon Tile
247           Container(
248             width: 64,
249             height: 64,

```

```

251       decoration: BoxDecoration(
252         color: product.themeColor.withOpacity(0.12),
253         borderRadius: BorderRadius.circular(16),
254       ),
255       child: Icon(
256         product.icon,
257         color: product.themeColor,
258         size: 30,
259       ),
260     ),
261     const SizedBox(width: 16),
262
263     // Name & Category
264     Expanded(
265       child: Column(
266         crossAxisAlignment: CrossAxisAlignment.start,
267         children: [
268           Text(
269             product.name,
270             style: const TextStyle(
271               fontSize: 16,
272               fontWeight: FontWeight.bold,
273               color: Color(0xFF2D3436),
274             ),
275           ),
276           const SizedBox(height: 4),
277           Container(
278             padding: const EdgeInsets.symmetric(
279               horizontal: 8, vertical: 2),
280             decoration: BoxDecoration(
281               color: Colors.grey.shade100,
282               borderRadius: BorderRadius.circular(6),
283             ),
284             child: Text(
285               product.category,
286               style: TextStyle(
287                 fontSize: 11,
288                 color: Colors.grey.shade600,
289                 fontWeight: FontWeight.w500,
290               ),
291             ),
292           ),
293         ],
294       ),
295     ),
296
297     // Price Tag (in ₹)
298     Column(
299       crossAxisAlignment: CrossAxisAlignment.end,
300       children: [

```

```

301         Text(
302           '₹${product.price.toStringAsFixed(0)}',
303           style: const TextStyle(
304             fontSize: 16,
305             fontWeight: FontWeight.w800,
306             color: Color(0xFF6C5CE7),
307           ),
308         ),
309         const SizedBox(height: 6),
310         const Icon(
311           Icons.arrow_forward_ios_rounded,
312           size: 14,
313           color: Colors.grey,
314         ),
315       ],
316     ),
317   ],
318 ),
319 );
320
321 }
322
323 // Modern Empty State Graphic
324 Widget _buildEmptyState() {
325   return Center(
326     child: Column(
327       mainAxisAlignment: MainAxisAlignment.center,
328       children: [
329         Container(
330           padding: const EdgeInsets.all(20),
331           decoration: BoxDecoration(
332             color: const Color(0xFF6C5CE7).withOpacity(0.1),
333             shape: BoxShape.circle,
334           ),
335           child: const Icon(
336             Icons.search_off_rounded,
337             size: 48,
338             color: Color(0xFF6C5CE7),
339           ),
340         ),
341         const SizedBox(height: 16),
342         const Text(
343           'No results found',
344           style: TextStyle(
345             fontSize: 18,
346             fontWeight: FontWeight.bold,
347             color: Color(0xFF2D3436),
348           ),
349         ),
350         const SizedBox(height: 6),
351         Text(
352           'Try adjusting your search or filters',
353           style: TextStyle(fontSize: 14, color: Colors.grey.shade500),
354         ),
355       ],
356     ),
357   );
358 }
359 }

```

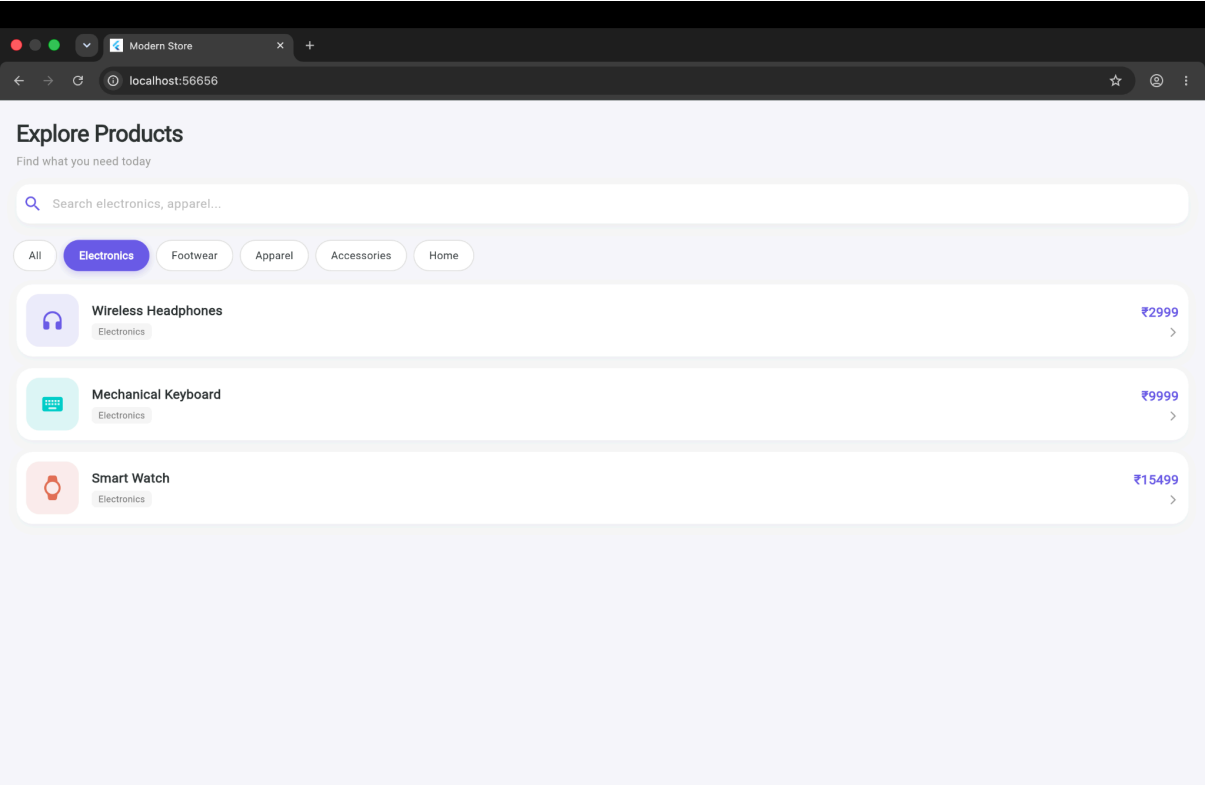
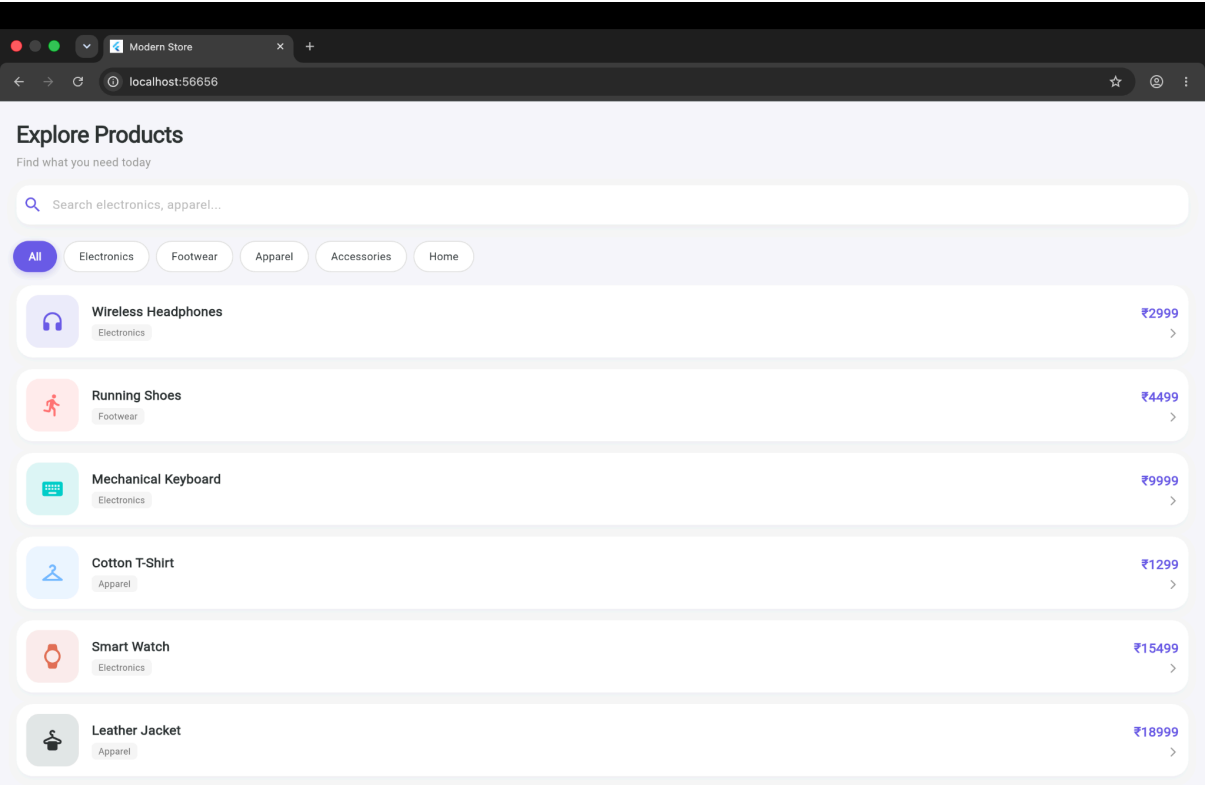
4. UI/UX Implementation Details

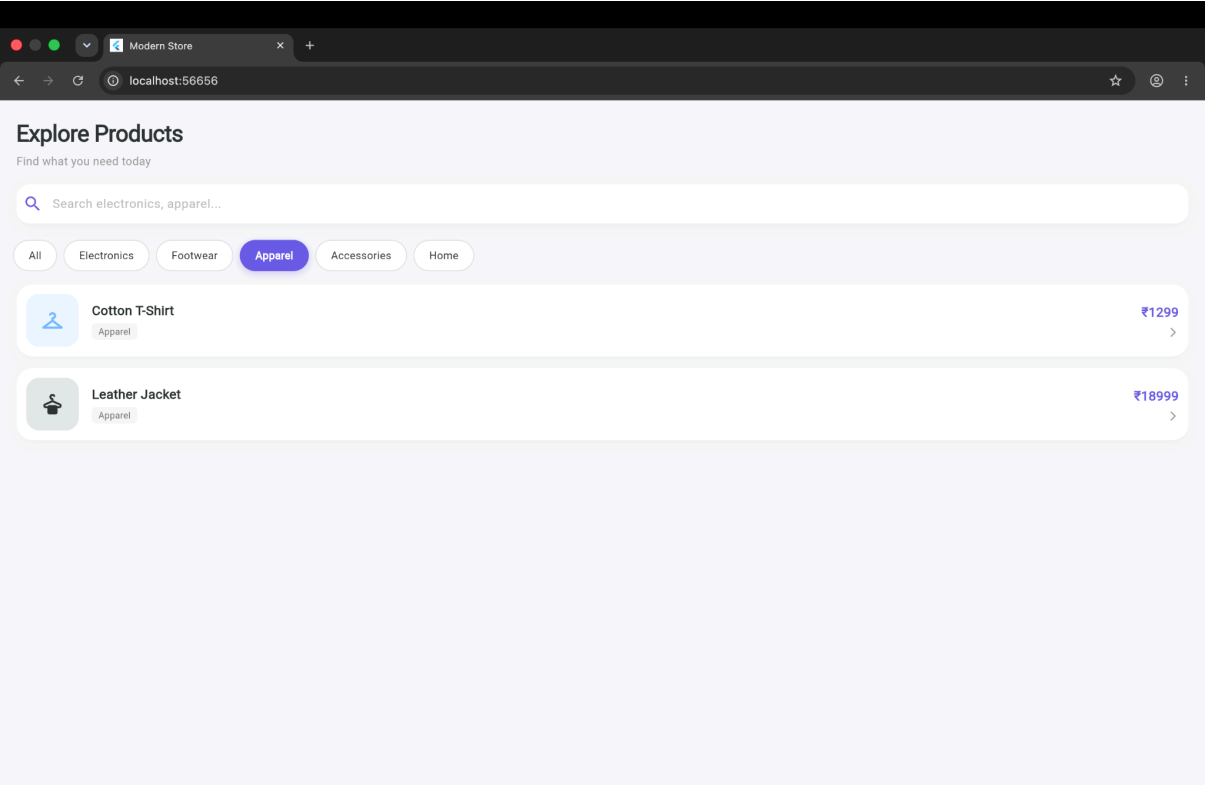
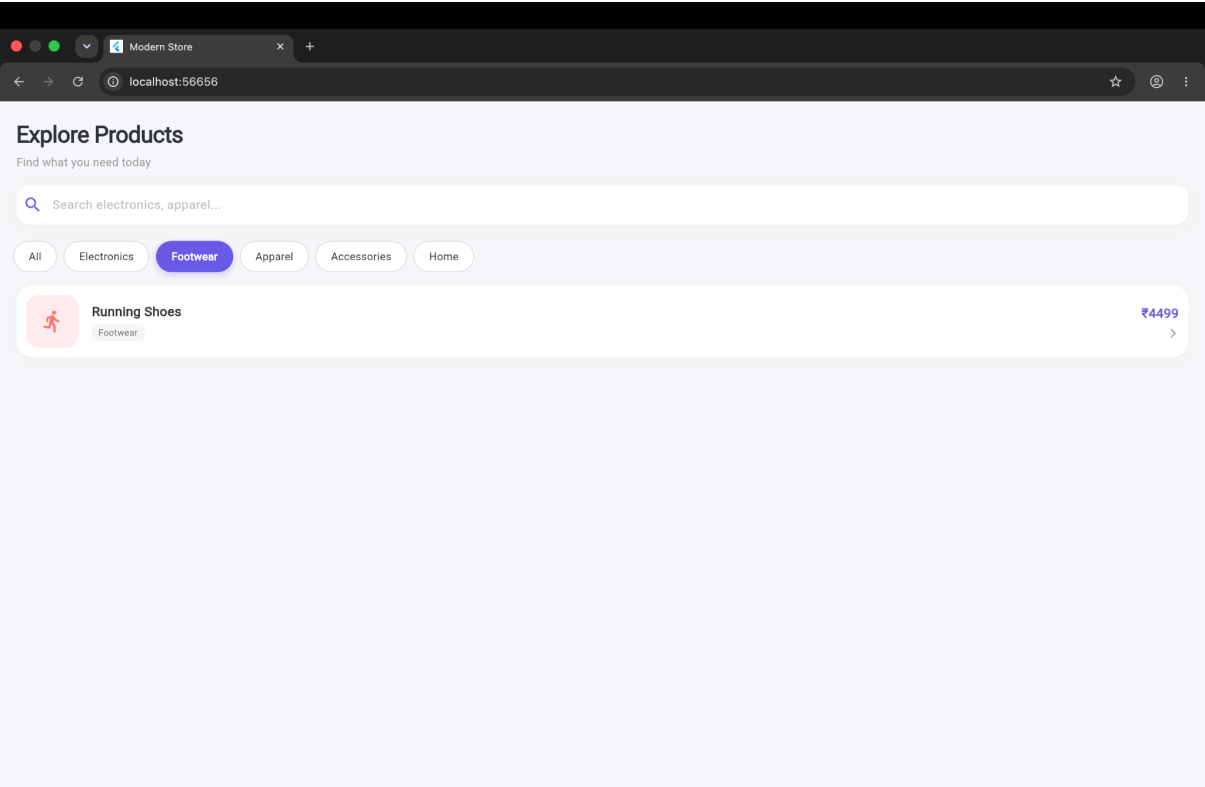
The user interface leans heavily on standard Flutter widgets, customized to create a premium feel.

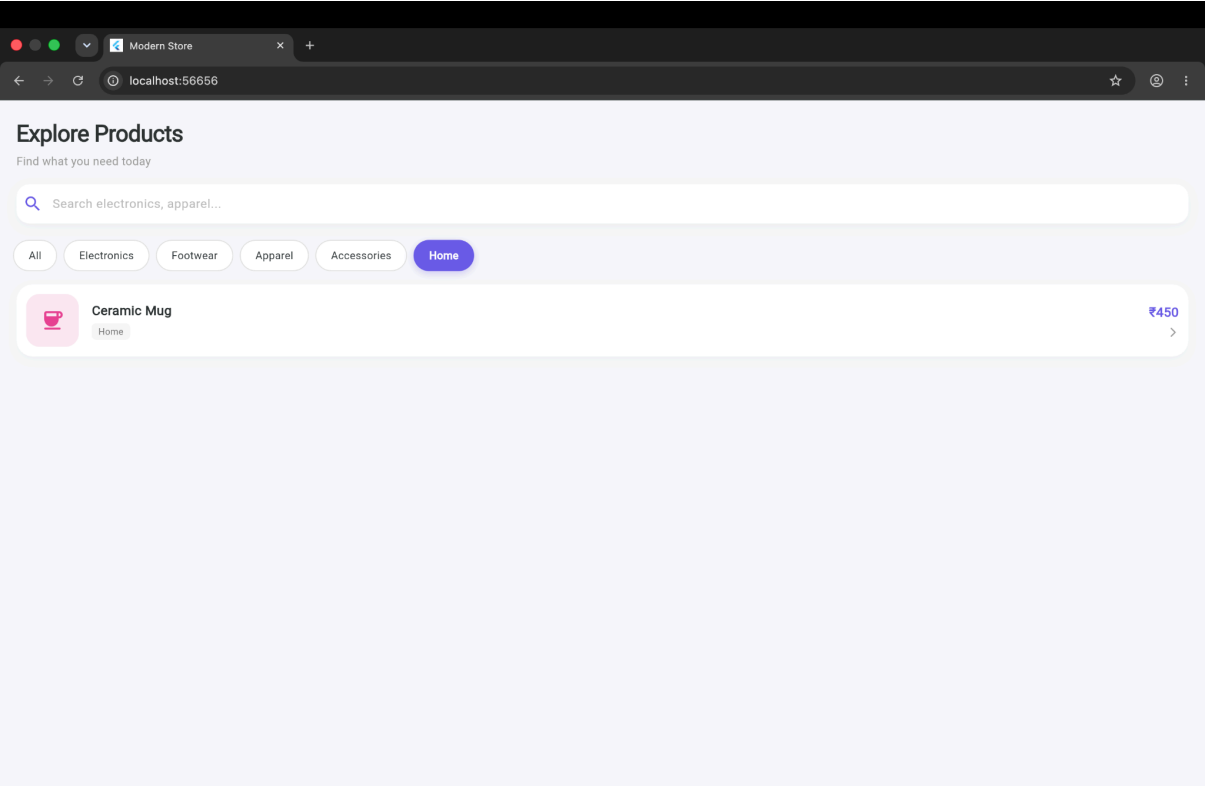
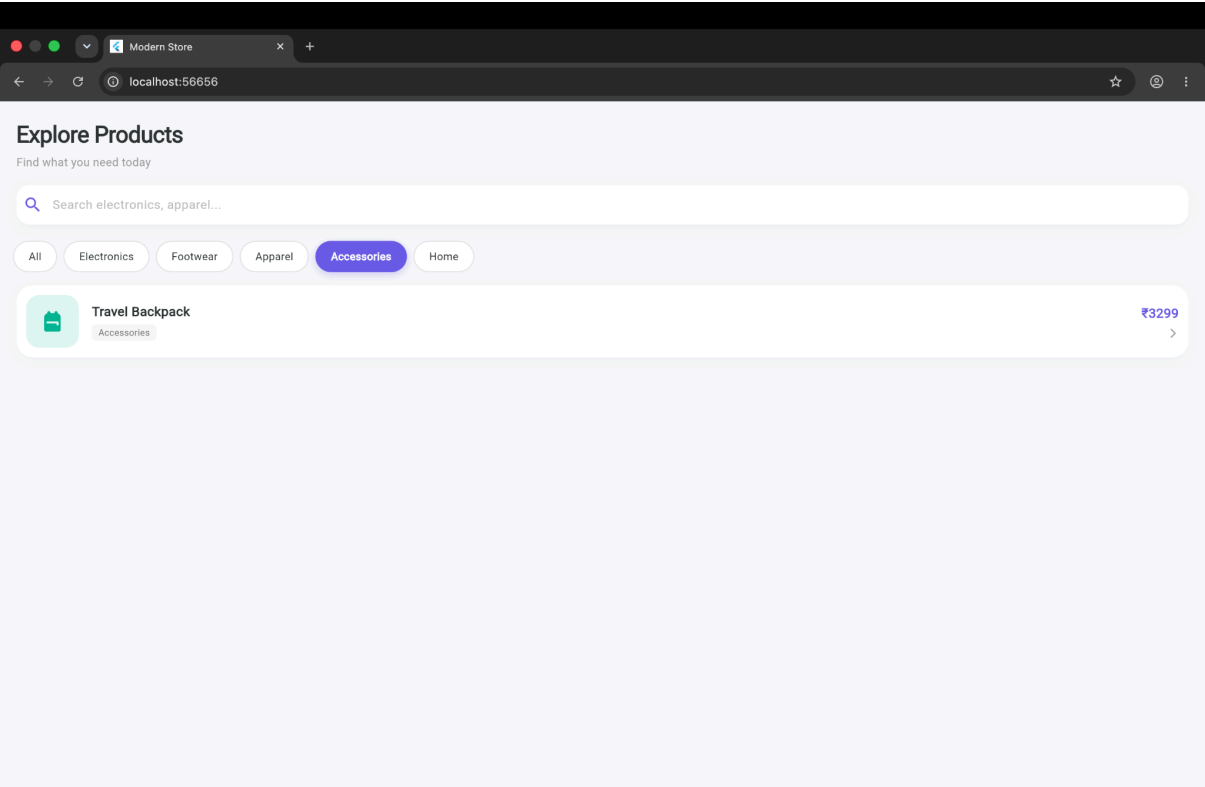
Widget	Application in Project
ListView.builder	Powers both the horizontal scrolling category chips and the vertical scrolling product list, ensuring memory efficiency for large lists.
TextField	Captures search queries in real-time. It includes custom shadow styling, an integrated search icon, and a conditional clear button.
AnimatedContainer	Provides smooth micro-interactions (color and shadow transitions) when a user taps a category chip.
Expanded	Prevents layout overflow by dynamically restricting the scrollable product list to the remaining safe vertical screen space.

Visual Feedback

- **Currency:** Prices are cleanly formatted using the Indian Rupee symbol (₹) without trailing decimals.
- **Empty States:** If a search yields zero results, the standard list is replaced by a visually distinct "Empty State" graphic (`_buildEmptyState()`) guiding the user to adjust their parameters.







5. Development Challenges & Resolutions

1. Memory Management:

- *Challenge:* The TextEditingController remained active in memory after the widget was destroyed.
- *Solution:* Hooked into the widget lifecycle by overriding the dispose() method to explicitly clear the controller.

2. Dynamic Data Extraction:

- *Challenge:* Hardcoding the category chips would lead to UI desync if the dataset changed.
- *Solution:* Utilized Dart's set mapping (...{for (var p in sampleProducts) p.category}) to automatically generate a unique list of categories directly from the active dataset.

3. Layout Constraints:

- *Challenge:* Placing a ListView directly inside a Column resulted in unbounded height errors.
- *Solution:* Wrapped the list in an Expanded widget to force it to respect the parent boundaries.

6. Technical Takeaways

- **Reactive UI:** Mastered the flow of user input mutating state variables, which in turn redraw the UI via setState().
- **Collection Virtualization:** Gained practical experience handling rendering performance using .builder constructors for dynamic lists.
- **Declarative Layouts:** Improved proficiency in stacking and styling standard widgets (Container, Row, Column) to build custom, modern interface cards.